TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Refractive Index of a Liquid Using a Plane Mirror and a Convex Lens

A film of liquid under the lens weakens its lower surface's refraction — the self-conjugate focus moves outward, and from that shift and the surface's radius, μ of the liquid.

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1. Aim and theory

Aim: to find the refractive index of a liquid (water in the model run). A convex lens rests on a plane mirror. A pin above the lens is raised or lowered until its image — formed by light passing through the lens, reflecting off the mirror normally, and returning through the lens — coincides with the pin itself (no parallax). At that setting the pin stands exactly at the focal point of the combination: the measured height is the focal length.

Dry, that height is F, the focal length of the glass lens alone. A few drops of liquid are then placed under the lens. The liquid fills the wedge between the lens's convex lower surface (radius R) and the mirror, and the light now crosses a **glass–liquid** interface at that curved surface instead of a glass–air one. Since the liquid's μ is much closer to glass's than air's is, refraction at the surface is **weaker**: the combination loses power, and the coincidence point moves **outward** to a longer focal length F′ > F. Treating the liquid layer as a thin (plano-concave) lens of focal length f_l used in contact with the glass lens:

$$1/F' = 1/F - 1/f_l \;\Rightarrow\; 1/f_l = 1/F - 1/F'$$

and the lensmaker formula for the liquid layer, whose one curved surface has radius R, gives $1/f_l = (\mu - 1)/R$, so

$$\mu = 1 + R/f_l$$

R is measured with a spherometer on the lens's lower surface.

2. Apparatus and procedure

- **Apparatus:** convex lens, plane mirror, pin on a height-adjustable stand with scale, spherometer, the liquid (water), dropper.
- Measure R of the lens's lower surface with the spherometer ($R = l^2/(6h_s) + h_s/2$, from its leg spacing l and sagitta h_s).
- Lens on mirror (dry): find the no-parallax pin height → F.
- Introduce a few drops of liquid under the lens (no bubbles, a uniform film, none spilling over the edge); find the new coincidence height → F′ (expect it larger than F).
- Compute f_l, then μ. Repeat with a fresh film and average.

3. Worked calculation (model data)

RESULT

Spherometer: R of the lens's lower surface = 20.0 cm.

Without liquid: coincidence height F = 15.0 cm. With the water film: F′ = 20.0 cm.

Liquid layer: $1/f_l = 1/F - 1/F' = 1/15.0 - 1/20.0 = (4 - 3)/60 = 1/60$ → $f_l = 60.0$ cm.

$\mu = 1 + R/f_l = 1 + 20.0/60.0 = 1 + 0.333 =$ **1.333** — the refractive index of water (1.333 at room temperature), recovered from three length measurements and a spherometer reading.

4. Precautions

- The no-parallax setting must be exact: move the eye side to side; pin and image must move together. Bracket the setting from above and below and average.
- The liquid film must be thin, uniform and bubble-free, and must not flood over the lens's edge (an over-filled film changes the surface geometry).
- Use the same lens surface against the liquid as the one whose R was measured.
- The mirror must be a true plane mirror, flat on the bench, and the lens must sit stably without rocking.
- Work promptly, and repeat the wet setting: evaporation shrinks the film and the coincidence height drifts as it does.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does the pin coincide with its image exactly when it stands at the focal point of the combination?
2. With F = 18.0 cm, F′ = 24.0 cm and R = 30.0 cm, compute μ of the liquid.

3. Why does adding the liquid lengthen the combination's focal length — the opposite of what a student who thinks "liquid lens = extra converging lens" expects?
4. What shape is the liquid layer, and which surface supplies the R in the formula?
5. Why is a spherometer needed at all — why can R not be taken from the glass lens's focal length F?
6. A bubble is trapped in the film. What error does it introduce, and in which direction?

Answer key

1. At the focal point, rays from the pin emerge from the lens parallel to the axis, strike the plane mirror at normal incidence, and retrace their paths exactly, refocusing at the pin itself. Only at the focus does the return image form at the object's own position, so coincidence locates the focal point with no screen and no image-distance reading.
2. $1/f_l = 1/18.0 - 1/24.0 = (4 - 3)/72 = 1/72 \rightarrow f_l = 72.0$ cm. $\mu = 1 + 30.0/72.0 = 1 + 0.417 = 1.417$ — a liquid more refractive than water.
3. Because the liquid does not add a surface; it changes one. Dry, the lens's lower surface refracts glass → air, a large μ step (1.50 → 1.00) and a strong refraction. Wet, the step is glass → liquid (1.50 → 1.33), much smaller, so that surface bends light less and the whole combination weakens — $F' > F$. The "extra lens" intuition fails because it imagines the liquid surrounded by air, which it is not.
4. The layer is plano-concave in shape — zero thickness at the centre where the lens touches the mirror, thickening toward the edge, flat beneath. Its curved boundary is the mould of the glass lens's lower surface, and that surface's radius R is the one in the formula; the plane side contributes no refraction.
5. Because F is the focal length of the whole glass lens, fixed by both of its surfaces and the glass's μ together; the liquid formula needs the radius of the one surface the liquid touches. The spherometer measures that geometry directly, independent of the glass lens's optics.
6. A bubble replaces liquid by air over part of the aperture: there the strong glass–air refraction returns, the weakening is locally undone, f_l comes out too short and μ too large. It also scatters light and roughens the image — the visible warning to remake the film.
